

Optional: Advanced MVC in Laravel

Organize routes and make errors predictable.

Learning Goals

- Group routes with a prefix and with `auth:sanctum` middleware.
- Avoid ambiguous route paths by ordering routes and adding constraints.
- Choose and document status codes for created, missing, invalid, and unauthenticated cases.

Group Related Routes

In `routes/api.php`, with `Route` and controllers imported:

```
Route::prefix('v1')->group(function () {  
    Route::apiResource('students', StudentController::class);  
    Route::apiResource('courses', CourseController::class);  
});
```

With the default API prefix, URLs start with `/api/v1/`.

Do not add `/api` twice. Implement each registered controller action.

Protect a Group

After completing the Sanctum setup from lesson 5:

```
Route::middleware('auth:sanctum')->group(function () {  
    Route::apiResource('students', StudentController::class);  
});
```

This protects every route in the group.

For mixed public/private access, protect individual routes instead.

Avoid Ambiguous Paths

```
Route::get('/students/search', [StudentController::class, 'search']);  
Route::get('/students/{student}', [StudentController::class, 'show'])  
    ->whereNumber('student');
```

Place specific routes before a general parameter route.

A numeric constraint prevents `search` from becoming a student ID.

A failed route constraint returns **404**.

Validate Only the Data You Save

Inside a controller method with a Request parameter:

```
$data = $request->validate([
    'name' => 'required|string|max:255',
    'email' => 'required|email',
]);
```

Use the returned `$data` for the fields you persist.

Request JSON errors with `Accept: application/json`.

Laravel validation errors normally return **422** for JSON requests.

Make the Contract Explicit

Case	Typical response in these Laravel examples
Record created	201
Missing model-bound record	404
Invalid validated input	422
Missing or invalid Bearer Token	401

Module 1 used 400 for several invalid inputs. Decide and document

which behavior your port should preserve or change.

Optional Practice

- Add a `v1` route group and inspect `route:list`.
- Test a specific route and a numeric record route.
- Test validation and authentication failures.
- Explain each status code in your tests.

Keep the basic request flow clear as the API grows.